1

Video Decomposition

*Integrating these three models yields **the final objective function** (**non-convex optimization**):*

$$\min_{\Theta} \mathcal{L}(\Theta) = \| \mathcal{H}^{\perp} \circ (\mathcal{X} - \mathrm{Fold}(UV^T) - \mathcal{R}) \|_F^2 + \| \mathcal{H} \circ (\mathcal{X} - \mathcal{F} - \mathcal{R}) \|_F^2 + \lambda \| \mathcal{F} \|_{TV} + \alpha \| \mathcal{H} \|_{3DTV} + \beta \| \mathcal{H} \|_1 + b \sum_{k=1}^{K} \sum_{s=1}^{n_k} \| \mathcal{M}_{ks} \|_1$$

$$s.t. \quad \mathcal{R} = \sum_{k=1}^{K} \sum_{s=1}^{n_k} D_{ks} \otimes \mathcal{M}_{ks}, \quad \| D_{ks} \|_F^2 < 1. \qquad (6)$$

***Regularization terms** to constrain the parameter space:*

Moving objects layer, relative to rain streak, **is more smooth**.

Sparsity of rain positions

Spatial-temporal continuity and **sparsity** of **moving objects**

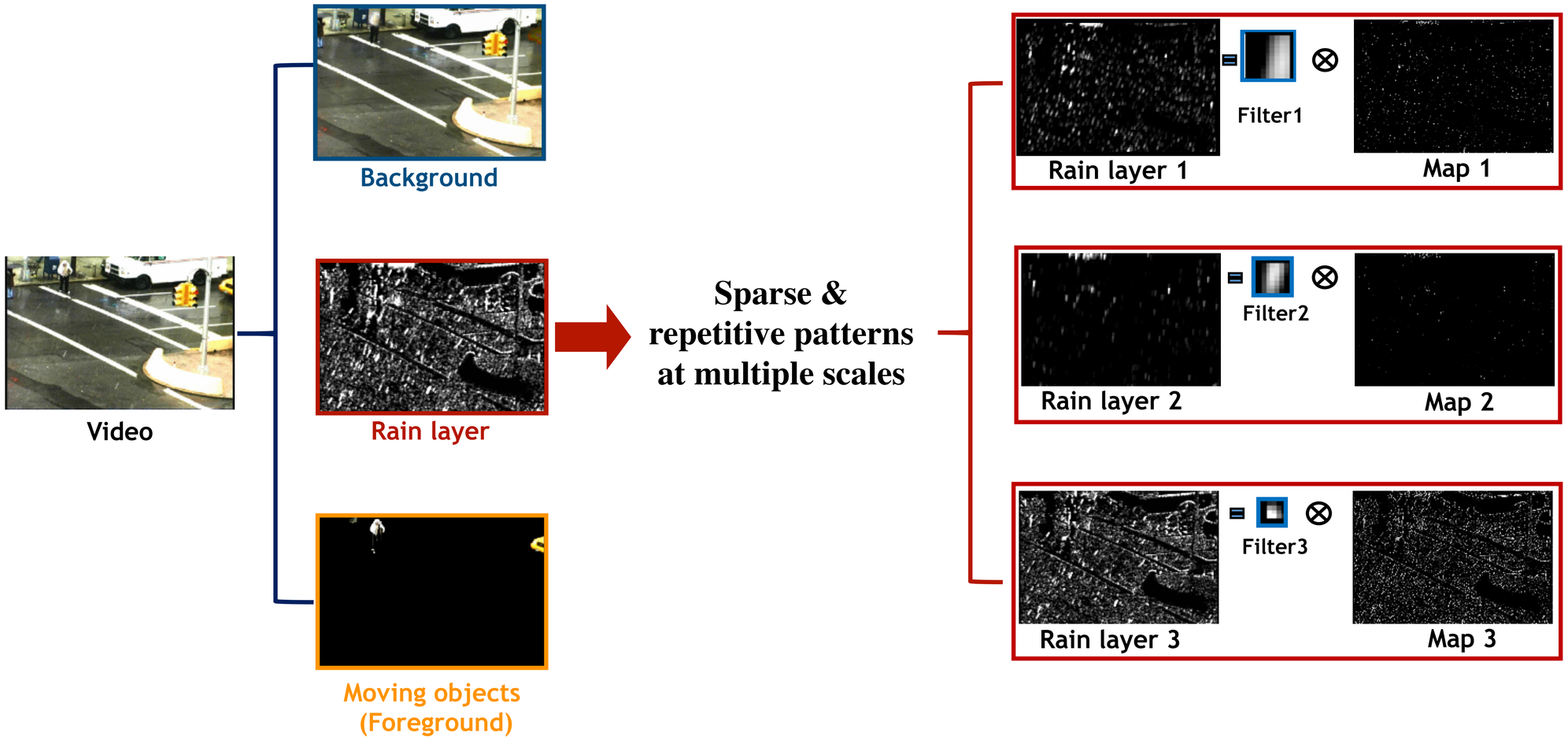
Alternating direction method of multipliers (ADMM) to iteratively optimize each variable.

OT-MSCSC: Online Rain/Snow Removal from Surveillance Videos (TIP 2021). **First author, high-impact, used in real-world environments**

1

Multi-scale Rain Layers

Align well with the statistical property of rain



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